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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: _____10_____

Team No.: _____25_____

Project Title: Disk Scheduling Algorithm Visualizer

Team Members

Roll Number	Student Name	Signature
2520090193	Sankeerthana Thanmaye	_____
2520090118	Vikram Immella	

1. Abstract

(Summarize the project idea, problem background, relevant Operating Systems/System Programming concepts, proposed Linux-based solution, and expected outcome. The abstract should not exceed 250 words.)

The Disk Scheduling Algorithm Visualizer is a web-based project that demonstrates how an operating system manages disk I/O requests. When multiple disk requests are waiting, the order in which they are served affects head movement and seek performance. This project visualizes four disk scheduling algorithms: First Come First Serve (FCFS), Shortest Seek Time First (SSTF), SCAN, and C-SCAN. The user enters the initial disk-head position, disk size, request queue, and direction where required. The application then calculates the service sequence, total head movement, and average seek distance, and presents the results using charts and a comparison table. The project applies core Operating Systems concepts such as disk I/O scheduling, request queues, seek time, and algorithm comparison. It is designed to run as a web application in a Linux environment using HTML, CSS, JavaScript, and Chart.js. The expected outcome is an interactive learning tool that helps users understand and compare the behaviour and performance of different disk scheduling algorithms.

2. Problem Statement

(Clearly describe the problem that the project intends to address and its relevance to Operating Systems and Systems Programming.)

In an operating system, several processes may generate disk I/O requests at the same time. Serving these requests in an inefficient order can cause unnecessary disk-head movement and increase seek time, which may reduce system performance. The project addresses this problem by implementing and comparing FCFS, SSTF, SCAN, and C-SCAN algorithms. It is relevant to Operating Systems because disk scheduling is an important part of I/O management and helps explain how the operating system organizes pending disk requests.

3. Objectives

(List the specific objectives of the project.)

1. Implement FCFS, SSTF, SCAN, and C-SCAN disk scheduling algorithms.

2. Accept the disk size, initial head position, request queue, and scan direction as input.

3. Display the disk-head service sequence and calculate total head movement and average seek distance.

4. Compare the algorithms using visual charts and a clear results table.

4. Proposed Methodology

(Briefly explain how the project will be designed and implemented in Linux. Describe the major functionality, approach, and flow of the proposed solution.)

The application will be developed using HTML, CSS, and JavaScript and tested in a Linux/Ubuntu environment. The user first enters the disk parameters and pending request queue. JavaScript validates the input and executes the selected scheduling algorithms. FCFS serves requests in arrival order; SSTF selects the nearest pending request; SCAN moves in one direction and then reverses; and C-SCAN moves in one direction and returns to the beginning before continuing. The application calculates the head movement for each algorithm and displays the service order, total movement, average seek distance, charts, and a comparison table using Chart.js.

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project
Disk I/O Scheduling	Organizes pending disk read/write requests.
FCFS	Services requests in the order they arrive.
SSTF	Selects the pending request closest to the current head position.
SCAN	Moves the head in one direction and then reverses direction.
C-SCAN	Services requests in one direction and returns to the beginning.
Seek Time / Head Movement	Measures and compares disk-head movement for each algorithm.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520090193	Sankeerthana Thanmaye	Responsible for project planning, algorithm implementation, web-page design.
2520090118	Vikram Immella	Responsible for testing, result visualization, documentation, and presentation.

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for development and testing.
C / C++	Not used in the current web-based implementation; can be left unused.
HTML, CSS, JavaScript	Build the user interface and implement the scheduling logic.
Chart.js	Display visual charts for algorithm comparison.
Visual Studio Code / Web Browser	Code editing, execution, and testing of the web application.

8. Expected Outcome

(Describe the expected functionality, output, and outcome of the project.)

The completed application is expected to accept disk scheduling inputs and correctly simulate FCFS, SSTF, SCAN, and C-SCAN. It will show the order in which requests are serviced, the movement of the disk head, total head movement, and average seek distance. Charts and a

comparison table will make the differences between the algorithms easy to understand. The project will serve as an interactive educational demonstration of disk I/O scheduling.

9. GitHub Repository

GitHub Repository Link: _____

Repository Created: _____

☐Yes ☐No

Faculty Name and Signature

Faculty Name: _____Dr. M. Raghupathi_____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____